BIG DATA:

According to Gartner, the definition of Big Data –

"Big data" is high-volume, velocity, and variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making."

This definition clearly answers the "What is Big Data?" -- Big Data refers to complex and large data sets that have to be processed and analyzed to uncover valuable information that can benefit businesses and organizations.

However, there are certain basic tenets (principles) of Big Data that will make it even simpler to answer what is Big Data:

- It refers to a massive amount of data that keeps on growing exponentially with time.
- It is so voluminous that it cannot be processed or analyzed using conventional data processing techniques.
- It includes data mining, data storage, data analysis, data sharing, and data visualization.
- The term is an all-comprehensive one including data, data frameworks, along with the tools and techniques used to process and analyze the data.

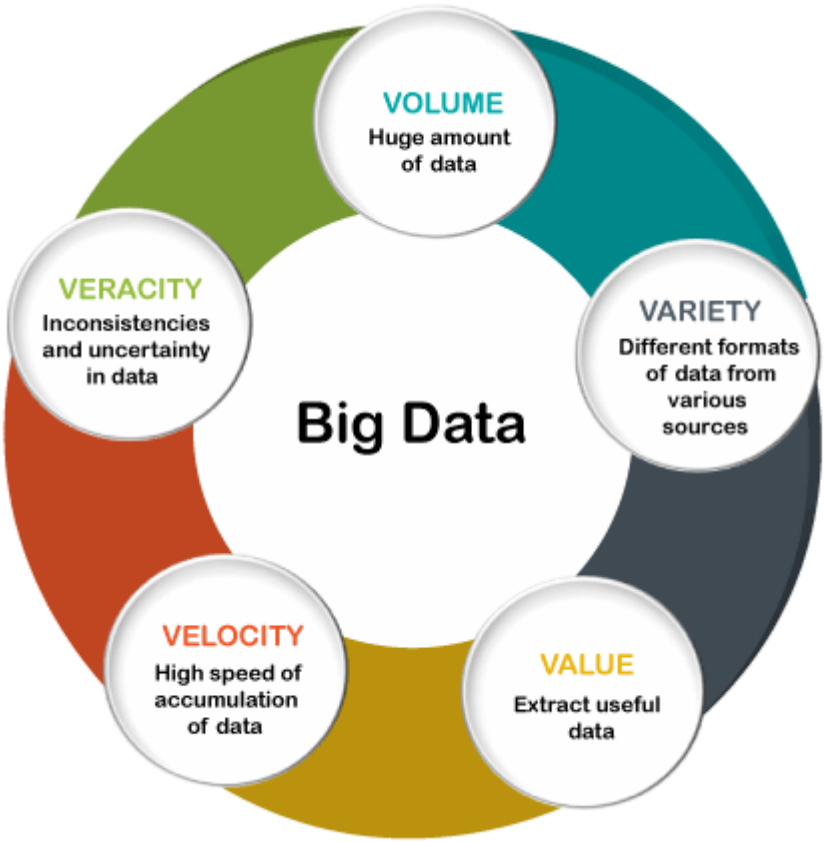
Big Data Characteristics

Big Data contains a large amount of data that is not being processed by traditional data storage or the processing unit. It is used by many **multinational companies** to **process** the data and business of many **organizations**. The data flow would exceed **150 exabytes** per day before replication.

There are five v's of Big Data that explains the characteristics.

5 V's of Big Data

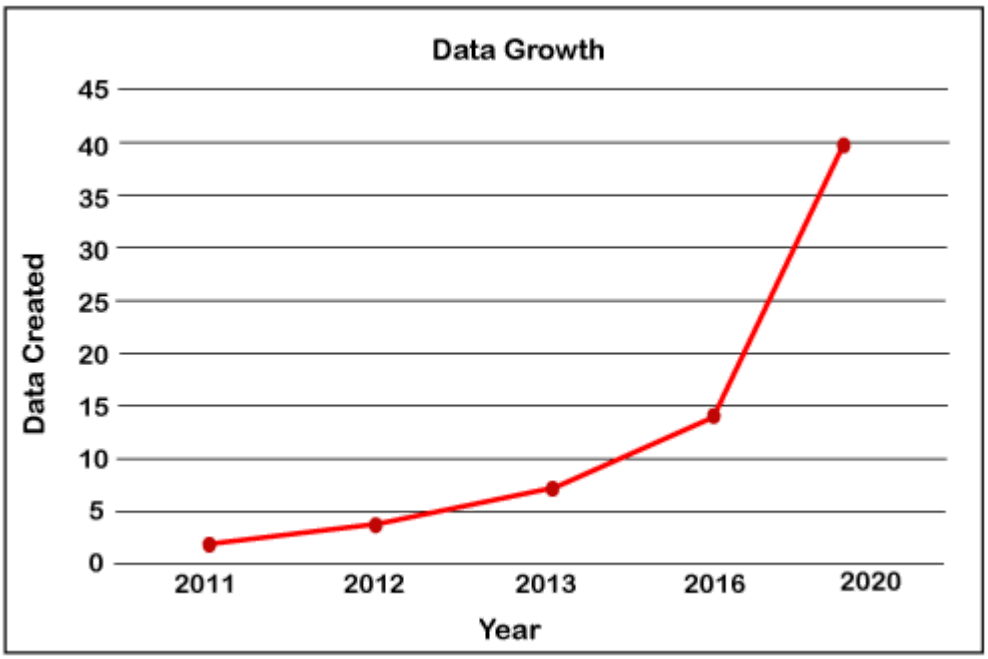
- **Volume**
- **Veracity**
- **Variety**
- **Value**
- **Velocity**



Volume

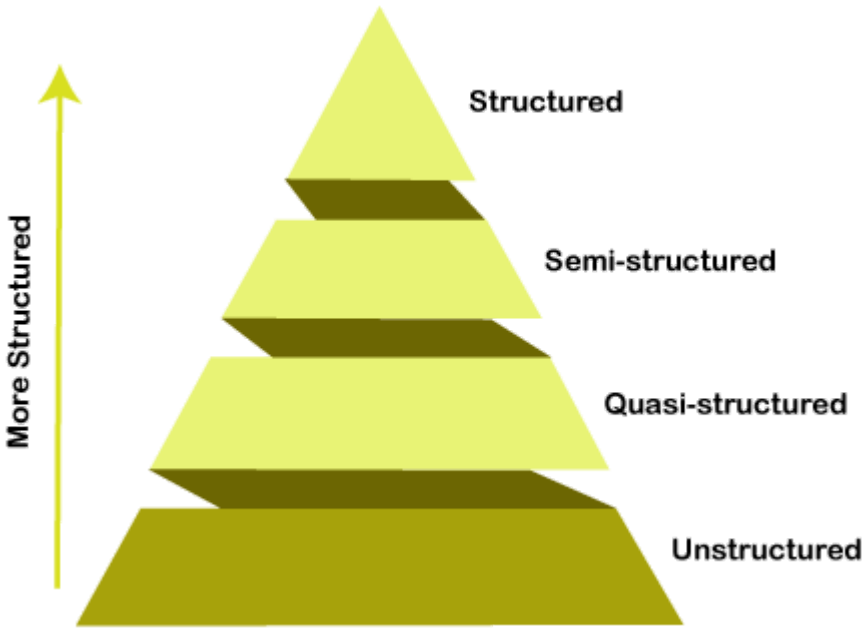
The name Big Data itself is related to an enormous size. Big Data is a vast 'volumes' of data generated from many sources daily, such as **business processes, machines, social media platforms, networks, human interactions,** and many more.

Facebook can generate approximately a **billion** messages, **4.5 billion** times that the "**Like**" button is recorded, and more than **350 million** new posts are uploaded each day. Big data technologies can handle large amounts of data.



Variety

Big Data can be **structured, unstructured, and semi-structured** that are being collected from different sources. Data will only be collected from **databases** and **sheets** in the past, But these days the data will comes in array forms, that are **PDFs, Emails, audios, SM posts, photos, videos,** etc.



The data is categorized as below:

a. **Structured data:** In Structured schema, along with all the required columns. It is in a tabular form. Structured Data is stored in the relational database management system.

a. **Semi-structured:** In Semi-structured, the schema is not appropriately defined, e.g., **JSON, XML, CSV, TSV**, and **email**. OLTP (**Online Transaction Processing**) systems are built to work with semi-structured data. It is stored in relations, i.e., **tables**.

a. **Unstructured Data**: All the **unstructured files, log files, audio files**, and **image** files are included in the unstructured data. Some organizations have much data available, but they did not know how to **derive** the value of data since the data is raw.

a. **Quasi-structured Data:**The data format contains textual data with inconsistent data formats that are formatted with effort and time with some tools.

Example: Web server logs, i.e., the log file is created and maintained by some server that contains a list of **activities**.

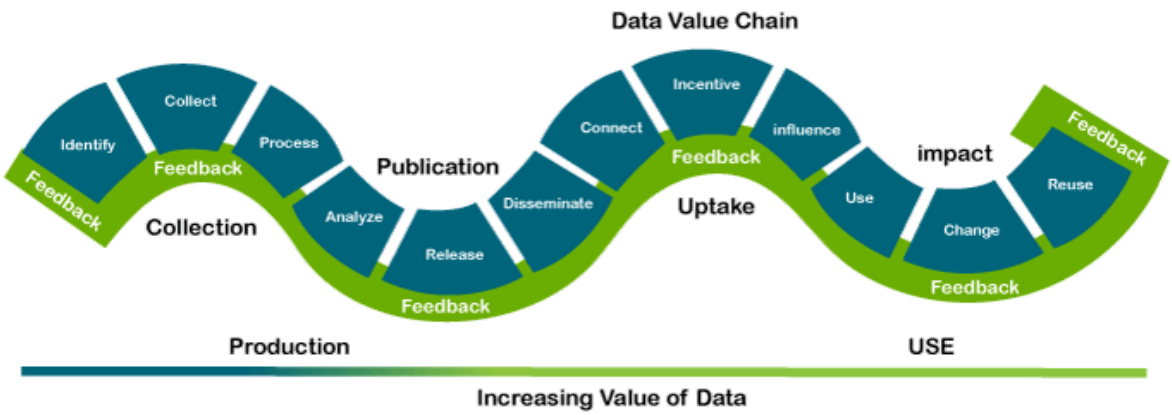
Veracity

Veracity means how much the data is reliable. It has many ways to filter or translate the data. Veracity is the process of being able to handle and manage data efficiently. Big Data is also essential in business development.

For example, **Facebook posts** with hashtags.

Value

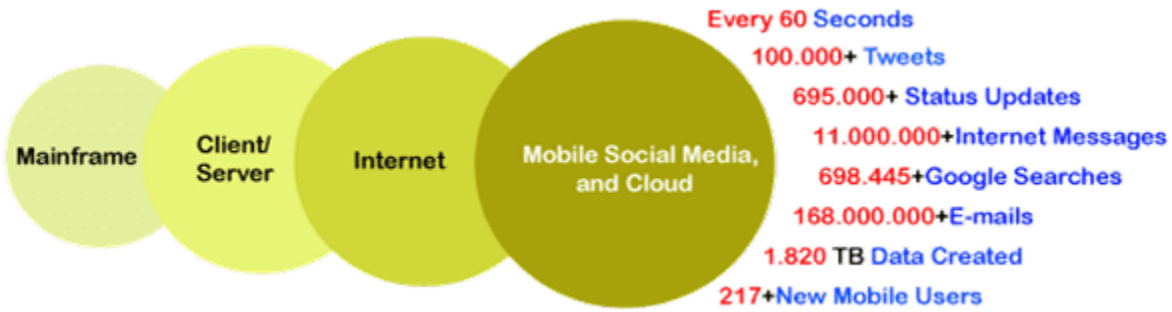
Value is an essential characteristic of big data. It is not the data that we process or store. It is **valuable** and **reliable** data that we **store, process**, and also **analyze**.



Velocity

Velocity plays an important role compared to others. Velocity creates the speed by which the data is created in **real-time**. It contains the linking of incoming **data sets speeds, rate of change**, and **activity bursts**. The primary aspect of Big Data is to provide demanding data rapidly.

Big data velocity deals with the speed at the data flows from sources like **application logs, business processes, networks, and social media sites, sensors, mobile devices,** etc.



What is a big data platform?

Big data platforms are comprehensive frameworks that enable organizations to store, process, and analyze vast amounts of structured and unstructured data.

Big data platform features

Big data platforms offer several features - from data sourcing to advanced analytics, helping businesses utilize data to achieve their business objectives. Some prominent features offered by big data platforms include:

a. Data storage and management

Data storage and management is a fundamental feature of big data platforms. These platforms provide robust and scalable storage solutions for handling large volumes of structured and unstructured data. They offer various storage options, such as distributed file systems, NoSQL databases, and data lakes, allowing organizations to store and organize data efficiently.

One key advantage of big data platforms is the support for distributed file systems like Hadoop Distributed File System (HDFS) and cloud-based storage options, which enable

seamless data storage across various environments. With advanced data management capabilities, these platforms facilitate data integration, cleansing, and transformation, ensuring the data is readily accessible for analysis and decision-making.

b. Distributed processing

Distributed processing is a crucial feature of big data platforms that enables processing large volumes of data across multiple nodes or servers in a distributed computing environment. This feature allows big data platforms to scale horizontally, meaning they can be easily expanded to handle more data by adding more nodes.

This approach enables parallel or simultaneous processing, significantly reducing the time required for data analysis. By distributing the workload across multiple nodes, big data platforms handle massive data sets that would be impractical on a single machine. Distributed processing is essential for achieving scalability and high performance in big data environments.

c. Fault tolerance

Fault tolerance refers to the ability of a system to continue functioning even in the event of software or hardware failures. The risk of failures is significantly higher in big data, where massive data is processed and analyzed. A fault-tolerant big data platform ensures that data processing and analytics operations can continue seamlessly, even if individual components or nodes within the system fail.

Fault tolerance is achieved through various techniques such as data replication, distributed computing, and automatic failover mechanisms. In the event of a hardware failure or software glitch, the system can seamlessly switch to backup resources, preventing data loss and minimizing disruptions. This feature ensures continuous data availability and uninterrupted processing crucial for mission-critical applications and real-time analytics.

d. Data analytics and visualization

The big data analysis platforms offer robust tools and algorithms that can process large volumes of data in real-time or near real-time. Big data platforms support various analytical techniques - from descriptive analytics to predictive and prescriptive analytics for processing complex business data.

Additionally, big data platforms offer advanced visualization capabilities, allowing users to create interactive dashboards, charts, and graphs to convey insights in a visually appealing and easily understandable manner. These capabilities enhance data comprehension and facilitate effective communication across different teams and stakeholders.

How do big data platforms work?

Big data platforms follow a structured process to ensure companies can harness data to make informed decisions. This process involves the following steps:

a. Data collection

Data collection is the initial step in the operation of big data platforms. It systematically gathers data from various sources such as databases, social media, sensors, and other sources. The data is collected using various methods such as web scraping, data feeds, APIs, and data integration tools. The collected data is then stored in a centralized repository, often a data lake or a data warehouse, where it can be easily accessed and processed for further analysis.

b. Data storage

Once the data is collected, it must be stored for efficient retrieval and processing. Big data platforms typically utilize distributed storage systems that can handle large volumes of data. These systems include Hadoop Distributed File System (HDFS), Google Cloud Storage, or Amazon S3. This distributed storage architecture ensures high availability, fault tolerance, and scalability.

c. Data processing

Once the data is collected, it must be processed to extract valuable insights. This process involves various operations such as cleaning, transforming, and aggregating the data. The parallel processing capabilities of big data platforms, such as Apache Hadoop and Apache Spark, enable rapid computations and complex data transformations.

d. Data analysis

Data analysis involves examining and interpreting large volumes of data to extract meaningful insights and patterns. The analysis process includes using machine learning algorithms, data mining techniques, or visualization tools to better understand the information. The analysis results can then be used to make data-driven decisions, optimize processes, identify opportunities, or solve complex problems.

e. Data quality assurance

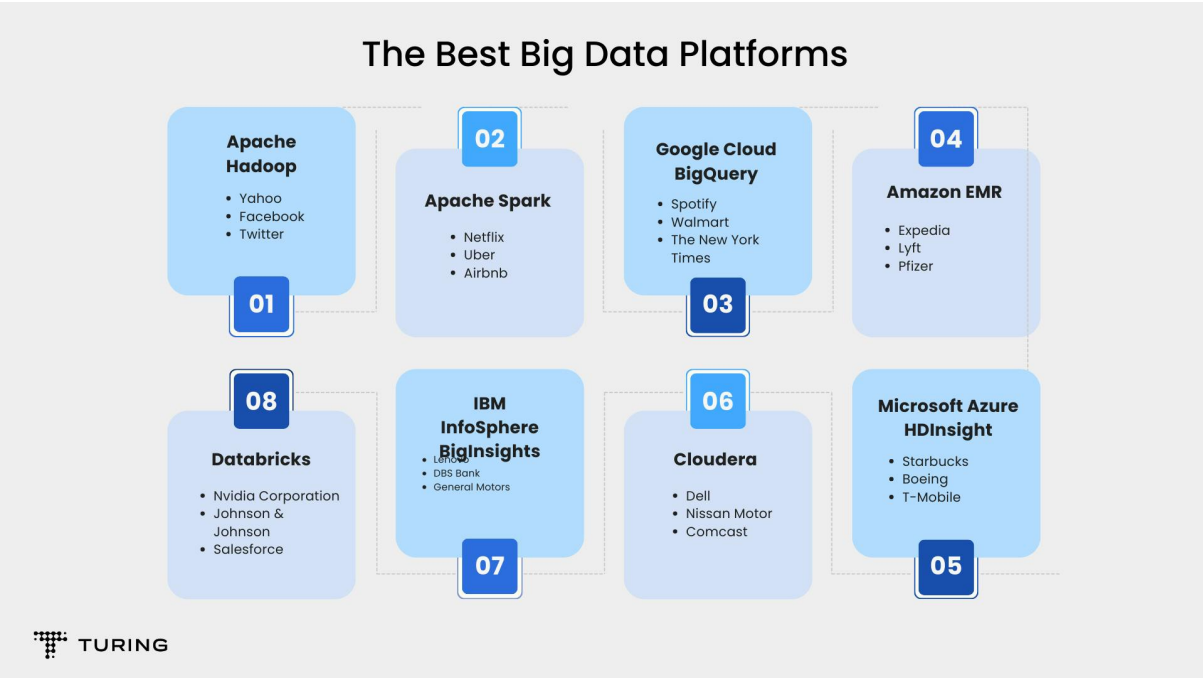
This stage ensures accuracy, consistency, integrity, relevance, and data security. The prominent techniques to implement data quality and governance include data quality management, lineage tracking, and cataloging. By implementing robust data quality assurance measures, organizations can have confidence in the data they use for decision-making.

f. Data management

Data management is a crucial aspect of big data platforms. It involves organizing, storing, and retrieving large volumes of data. Platforms employ various techniques such as data backup, recovery, and archiving to manage data effectively. These techniques help implement fault tolerance and ensure optimized data retrieval for all use cases.

The Big data platforms

Several big data platforms offer comprehensive features and solutions for businesses to manage and analyze complex datasets. The most prominent big data platforms used by companies include the following:



a. Apache Hadoop

Apache Hadoop is one of the industry's most widely used big data platforms. It is an open-source framework that enables distributed processing for massive datasets throughout clusters. Hadoop provides a scalable and cost-effective solution for storing, processing, and analyzing massive amounts of structured and unstructured data.

One of the key features of Hadoop is its distributed file system, known as Hadoop Distributed File System (HDFS). HDFS enables data to be stored across multiple machines, providing fault tolerance and high availability. This feature allows businesses to store and process data at a previously unattainable scale. Hadoop also includes a powerful processing engine called MapReduce, which allows for parallel data processing across the cluster. The prominent companies that use Apache Hadoop are:

- Yahoo
- Facebook
- Twitter

b. Apache Spark

Apache Spark is a unified analytics engine for batch processing, streaming data, machine learning, and graph processing. It is one of the most popular big data platforms used by

companies. One of the key benefits that Apache Spark offers is speed. It is designed to perform data processing tasks in-memory and achieve significantly faster processing times than traditional disk-based systems.

Spark also supports various programming languages, including Java, Scala, Python, and R, making it accessible to a wide range of developers. Hadoop offers a rich set of libraries and tools, such as Spark SQL for querying structured data, MLlib for machine learning, and GraphX for graph processing. Spark integrates well with other big data technologies, such as Hadoop, allowing companies to leverage their existing infrastructure. The prominent companies that use Apache Spark include:

- Netflix
- Uber
- Airbnb

c. Google Cloud BigQuery

Google Cloud BigQuery is a top-rated big data platform that provides a fully managed and serverless data warehouse solution. It offers a robust and scalable infrastructure for storing, querying, and analyzing massive datasets. BigQuery is designed to handle petabytes of data and allows users to run SQL queries on large datasets with impressive speed and efficiency.

BigQuery supports multiple data formats and integrates seamlessly with other Google Cloud services, such as Google Cloud Storage and Google Data Studio. BigQuery's unique architecture enables automatic scaling, ensuring users can process data quickly without worrying about infrastructure management. BigQuery offers a standard SQL interface for querying data, built-in machine learning algorithms for predictive analytics, and geospatial analysis capabilities. The prominent companies that use Google Cloud BigQuery are:

- Spotify
- Walmart
- The New York Times

d. Amazon EMR

Amazon EMR is a widely used big data platform from Amazon Web Services (AWS). It offers a scalable and cost-effective solution for processing and analyzing large datasets using popular open-source frameworks such as Apache Hadoop, Apache Spark, and Apache Hive. EMR allows users to quickly provision and manage clusters of virtual servers, known as instances, to process data in parallel.

EMR integrates seamlessly with other AWS services, such as Amazon S3 for data storage and Amazon Redshift for data warehousing, enabling a comprehensive big data ecosystem. Additionally, EMR supports various data processing frameworks and tools, making it suitable for a wide range of use cases, including data transformation, machine learning, log analysis, and real-time analytics. The prominent companies that use Amazon EMR are:

- Expedia
- Lyft
- Pfizer

e. Microsoft Azure HDInsight

Microsoft Azure HDInsight is a leading big data platform offered by Microsoft Azure. It provides a fully managed cloud service for processing and analyzing large datasets using popular open-source frameworks such as Apache Hadoop, Apache Spark, Apache Hive, and Apache HBase. HDInsight offers a scalable and reliable infrastructure that allows users to easily deploy and manage clusters.

HDInsight integrates seamlessly with other Azure services, such as Azure Data Lake Storage and Azure Synapse Analytics, offering a comprehensive ecosystem of Microsoft Azure services. HDInsight supports various programming languages, including Java, Python, and R, making it accessible to a wide range of users. The prominent companies that use Microsoft Azure HDInsight are:

- Starbucks
- Boeing
- T-Mobile

f. Cloudera

Cloudera is a leading big data platform that offers a comprehensive suite of tools and services designed to help organizations effectively manage and analyze large volumes of data. Cloudera's platform is built on Apache Hadoop, an open-source framework for distributed storage and processing of big data. Cloudera is a hybrid data platform deployed across on-premise, cloud, and edge environments.

Cloudera offers a unified platform that integrates various components such as Hadoop Distributed File System (HDFS), Apache Spark, and Apache Hive, enabling users to perform various data processing and analytics tasks. Cloudera also provides machine learning and advanced analytics tools, allowing businesses to gain deeper insights from their data. The prominent companies that use Cloudera are:

- Dell
- Nissan Motor
- Comcast

g. IBM InfoSphere BigInsights

IBM InfoSphere BigInsights is a powerful big data platform that offers a range of tools to manage and analyze large volumes of structured as well as unstructured data in a reliable manner. IBM InfoSphere BigInsights can handle massive data, making it suitable for enterprises dealing with complex datasets. It provides a comprehensive set of features for data management, data warehousing, data analytics, machine learning, and more.

IBM InfoSphere BigInsights provides a user-friendly interface and intuitive data exploration and visualization tools. The platform also offers robust security and governance features, ensuring data privacy and compliance with regulatory requirements. BigInsights is built on top of Apache Hadoop and Apache Spark, and it integrates with other IBM products and services, such as IBM DB2, IBM SPSS Modeler, and IBM Watson Analytics. This

integration makes it a good choice for businesses already using the IBM product/services ecosystem. The prominent companies that use IBM Infosphere BigInsights are:

- Lenovo
- DBS Bank
- General Motors

h. Databricks

Databricks is a prominent big data platform built on Apache Spark. Databricks simplifies the process of building and deploying big data applications by providing a scalable and fully managed infrastructure. It allows users to process large datasets in real-time, perform complex analytics, and build machine learning models using Spark's powerful capabilities.

Databricks provides an interactive workspace where users can write code, visualize data, and collaborate on projects. It also integrates with popular data sources and tools, making it easy to ingest and process data from various sources. With its auto-scaling capabilities, Databricks ensures that users have the resources to handle their workloads efficiently. Its automated infrastructure management and scaling capabilities make it a reliable choice for handling large datasets and complex workloads. The prominent companies that use Databricks are:

- Nvidia Corporation
- Johnson & Johnson
- Salesforce